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Readability assessment in Afrikaans: Cloze scores, linguistic features and cross-validated linear regression

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Abstract

Readability formulas offer a simple way to assess text difficulty by harnessing the relationship between the established difficulty of a set of texts and linguistic features of those texts. This study aimed at developing a readability formula for Afrikaans, using regression analysis, an approach applied to develop readability formulas traditionally. The readability of 40 Afrikaans texts was established via cloze tests, and 88 linguistic features were extracted automatically from those texts. Linear regression with four-fold cross-validation was performed using the cloze scores as dependent variable and linguistic features as predictors. Frequent predictors in the cross-validation formulas were word length and concreteness, both with a negative correlation with readability, and number of verb phrases, with a positive correlation. The difference between the results for the four cross-validation training subsets of 30 texts each (variance explained: 50%) and their counterpart test subsets of 10 texts each (variance explained: 18%) suggests limited generalisability to unseen data. Our findings underscore the importance of testing newly developed formulas on independent datasets to avoid overfitting. These insights may be valuable for future research on Afrikaans readability.

Introduction

The success of a written text hinges on several factors, one of which is its readability: the ease with which a text can be read and understood by its audience (Dale & Chall 1949; Klare 1963; McLaughlin 1969; DuBay 2004; Kleijn 2018). In an age inundated with written information, the readability of a text is crucial for capturing and retaining audience attention. Readability assessment, the process by which readability is measured, can help to determine whether written information is sufficiently accessible to its intended audiences, and can assist authors to ensure clear communication with their target readers.

The importance of the readability of written public information is underscored by legislation passed all over the world. In South Africa both the National Credit Act, 34 of 2005, (RSA 2006) and the Consumer Protection Act, 68 of 2008, (RSA 2009), require the use of plain language to make public information more readable and accessible. Similarly, the American Plain Writing Act, 111-274 of 2010 (USA 2010), and various European regulations highlight the necessity for clear and comprehensible written public information (Cornelius 2015; Kleijn 2018; Clarity International 2022).

Readability assessment is particularly useful for categorising reading material to match a reader's level or ability. Readers benefitting from reading-level distinctions or having texts matched with their abilities or expertise include school learners, university students and learners of an additional language (Vajjala & Meurers 2012; Crossley et al. 2017).

As Sibeko and Van Zaanen (2022) point out, readability assessment can be approached from different perspectives, which may involve reader characteristics, text characteristics (for example, syntactical and lexical features) and/or text presentation (layout and typography), to name a few. This study focuses on a frequently used instrument for readability assessment, the readability formula, involving text characteristics.

Readability formulas harness correlations between certain text characteristics and text difficulty to give users an indication of the readability of their particular texts. In this article, we touch on the development of traditional readability formulas in general and then present an exploratory study aimed at developing a novel readability formula for Afrikaans, a non-agglutinative Indo-European Germanic language spoken in southern Africa, and one of the twelve South African official languages.

Aim of the current study

There is no readability formula for Afrikaans of which the usefulness has been demonstrated by empirical research, to the authors' knowledge. VIVA (the Virtual Institute for Afrikaans) has recently made available a web-based platform to explore the readability of Afrikaans texts (VIVA 2024) incorporating Combrink (1992)'s *Misindeks* and a formula developed for Afrikaans by McDermid Heyns (2007), as well as two well-known readability formulas for English, the Reading Ease formula (Flesch 1948) and the Flesch-Kincaid formula (Kincaid et al. 1975). These adapted means of readability assessment have, however, not been evaluated for general Afrikaans texts by their developers. Moreover, Jansen et al. (2017) have illustrated that the McDermid Heyns (2007) formula does not generalise well enough to be used for general Afrikaans texts. Other than the VIVA web platform there are no electronic readability assessment tools specifically developed for Afrikaans.

The aim of this study is to develop a novel readability formula for Afrikaans, based on research on the relationship between automatically measurable linguistic features of Afrikaans texts, on the one hand, and actual comprehension data of those texts by adult Afrikaans-speaking readers, on the other hand. We follow a strict evaluation methodology in which we require that developed formulas are tested on texts that have not been used to develop those formulas to establish whether they can be expected to work well on new, unseen, texts. A formula that generalises well could be incorporated into digital tools that allow writers of Afrikaans texts to predict how accessible their text would be to a wide adult readership.

Traditional readability formulas

Interest in measuring text success dates back to the late 19th century, with Sherman (1883) finding that sentence lengths had decreased by half since Shakespearean times, and that the average sentence length remained consistent among individual authors. His work initiated the classical period in readability; his findings laid the groundwork for using text samples to predict readability and introduced statistical analysis into literary criticism (DuBay 2004).

Vogel and Washburne (1928) were the first to develop a readability formula based on regression analysis. After eliminating factors that correlated highly with each other, the linguistic features they used in their formula were the percentage of unique words and

simple sentences, together with the number of prepositions and the variable “frequency according to the Thorndike list”. Lists of “difficult words”, usually based on frequency counts, have since become a frequent element in readability formulas.

By 1948 two formulas, still used today, emerged: the Flesch Reading Ease formula, aimed at adults, and the Dale-Chall formula, aimed at primary and secondary schoolchildren. Consistent with traditional readability formulas, both formulas are based on explicit linguistic features: a syntactic predictor and a lexical predictor. The Flesch Reading Ease formula (Flesch 1948) uses the average number of words per sentence and the average number of syllables per 100 words, whereas the Dale-Chall formula (Dale and Chall 1949) uses the average number of words per sentence and the number of words that are not in Dale’s word-frequency list. By 1975 the Flesch formula had been adapted for use in the US navy to become known as the Flesch-Kincaid Grade Level formula (Kincaid et al. 1975).

The Gunning-Fog Index, a readability formula developed for adults, was published in 1952. It uses two linguistic features: average sentence length and the number of words with more than two syllables per 100 words (Gunning 1952).

DuBay (2004) points out that the Flesch, Dale-Chall and Gunning-Fog Index formulas and research involving those traditional formulas brought readability to public attention. Their work stimulated new consumer demands for documents in plain language and further research into the readability of texts. The development of those formulas also concluded the classical period during which formulas were characterised by an understandable focus on simplicity, as all counts and calculations had to be done by hand.

During the 1960s, the cloze test revitalised readability research by providing a valid criterion for assessing readability, as shown in studies by Coleman (1965) and Bormuth (1968) reporting significantly higher correlation than earlier methods. Readability formulas started containing more predictors, made possible by the ability of computers to count text features. However, the focus on machine-countable predictors led to criticism, as more complex factors such as cohesion, coherence and the interactive nature of reading were often overlooked. As a response, new approaches emerged, incorporating psycholinguistic dimensions like inference load (Kintsch 1979; Kemper 1983) and conceptual density (Kintsch & Vipond 1979). Francois (2015) reflects that while these complex variables showed

potential, they were labour-intensive and yielded only modest performance gains over traditional formulas. Combining these cognitive features with classic predictors led to minimal improvements, suggesting that simpler readability models still hold practical value. The 1990s brought formulas the most prominent of which still relied on the traditional approach. The updated Chall and Dale (1995) formula, for example, is based on three features: number of frequent words, number of unfamiliar words, and number of sentences.

The late 1990s ushered in the AI readability paradigm, as François (2015) calls it. A readability system relying on natural language processing (NLP) software was introduced by Daoust et al. (1996), and Folz et al. (1998) used latent semantic analysis to investigate the lexical coherence of a text. Si and Callan (2001) were the first to approach the prediction of readability as a classification problem, with Collins-Thompson and Callan (2004) continuing that research by focusing on the lexical components of the model. Schwarm and Ostendorf (2005) use a support vector model (SVM) for readability prediction. The development of over the past 20 years, along with the availability of new sources of data from the internet and social media, has been an impetus for increased research into text readability assessment (see Collins-Thompson (2015), François (2015) and Vajjala (2022) for an overview of more recent developments). While further discussion is beyond the scope of this article, studies by Martinc et al. (2021) and Vajjala (2022) have shown that pre-trained neural language models like BERT (Devlin et al. 2019) can assess readability without the explicit inclusion of linguistic features.

Despite the advances in automatic readability assessment, traditional formulas are still used widely (Vajjala & Meurers 2014). The relative simplicity, ease of use and comparability of readability formulas within certain parameters make such formulas attractive to apply (Crossley et al. 2019; Ante 2022; McGrath, Millar & Moore 2022; Lee & Lee 2023). Marchisio et al. (2019) use the Flesch Reading Ease formula, the Flesch-Kincaid Grade Level formula and the Dale-Chall formula as measures of readability to control the readability level of machine translated text, for instance, and Microsoft Office uses the Flesch-Kincaid Grade Level formula and Flesch Reading Ease formula to determine readability levels for its users (Microsoft 2022). In a systematic review of the readability of publicly available mental health information, Boutemen and Miller (2023) found that among formulas used to determine readability of health information, the Flesch-Kincaid was used most frequently, appearing in

21 of 33 studies. Berninger et al. (2021) investigated the readability levels of articles in 16 finance journals published between 2000 and 2016 which all used Flesch-Kincaid readability scores.

Given the wide use of traditional readability formulas, the next section explores how they are generally developed.

The mechanics of a traditional readability formula

François (2015) summarises the development of a traditional readability formula into three key steps: obtaining an indication of the readability of a set of texts aimed at a specific audience, identifying linguistic features that predict readability in those texts, and combining those predictors in a formula to produce a readability index.

Step 1 in this process requires identifying a suitable set of texts aimed at a particular target audience, and then establishing a criterion that assesses the readability of each of the texts in the set to that target audience. Methods of assessment include comprehension tests, reading time, obtaining judgements from readers and administering cloze tests (Klare 1974; François 2015).

Step 2 involves identifying the linguistic features occurring in the set of texts that may influence the readability of those texts, and that may serve as predictors of the difficulty of the texts. The viability of a single feature as a possibly relevant variable is usually judged by considering its correlation[1] with the criterion representing the difficulty of the texts. If the feature is sufficiently correlated with the difficulty – positively or negatively – it may serve as a predictor in the readability formula.

Step 3 is to combine the various predictors in a formula that gives a single readability index as output. The following multiple regression equation would represent such a formula:

$$Y = B_0 + B_1 X_1 + B_2 X_2 + ... + B_n X_n$$

The coefficients B_1, [...], B_n, which represent the relative weight of each linguistic feature, are automatically inferred from the set of texts by means of the statistical least squares method.

In the current study, we included a fourth step, namely testing the newly developed formula to establish how well regression analysis with the selected predictors work, and whether the

formula can be applied to texts that were not used to develop the formula. Work done by Jansen et al. (2017), Kleijn (2018) and Pander Maat et al. (2023) underscores both the importance of testing newly developed readability formulas with unseen data, and the usefulness of cloze scores as readability criterion. Testing and evaluation is standard practice in machine learning (Goldberg 2017) – it is a way to establish whether a machine learning model generalises well on new data. Statistical modelling for the development of readability formulas may do well to follow this practice.

In the next section, cloze tests as a popular criterion in the development and evaluation of a readability formula are explored.

The role of cloze tests in the development of readability formulas

The introduction of the cloze test as a criterion for readability by Taylor (1953) made it possible to test the properties of texts and readers more accurately and in more detail than before. A cloze test is created by deleting every n^{th} word (usually every fifth word) of a text and replacing the deleted part by a blank space or a line, always of the same length, essentially creating gaps in the text. Cloze test participants are then asked to fill in those gaps by inserting the word they expect to have been there in the original text. The average percentage of correct answers for a sample of readers from the target audience is regarded as a measure of the comprehensibility of the original text for that audience.

Taylor (1953) proposed the cloze test for readability measurement, arguing that asking readers to fill in blanks at regular intervals in a cloze test captures the similarity between an author's writing patterns and the reader's anticipation (1953: 417). Several studies since have shown strong correlations between cloze scores and traditional comprehension test scores (Bormuth 1967, 1968; Kamalski 2007; Gellert & Elbro 2013). This is likely because cloze tasks mirror natural reading processes, which Goodman (1967: 126-127) describes as a "psycholinguistic guessing game": readers and cloze test participants alike rely on language knowledge, memory cues and contextual expectations (see also Lopes Rego et al. 2024). The cloze test's resemblance to reading confers strong construct validity (Horton 1974; Jansen et al. 2017), and its simplicity, reliability and ease of administration makes it a good alternative to traditional comprehension tests.

Cloze tests can be constructed and scored in several ways. With regard to its construction, a choice is to be made between the fixed-ratio method and the rational fill-in method when deciding which words to remove from the original texts. The fixed-ratio method involves replacing the nth word with a gap (e.g. the first, sixth, eleventh word, and so on), whereas the rational fill-in method allows researchers themselves to identify which kinds of words to leave out, for example, only verbs, nouns, adjectives, jargon and/or prepositions (Kobayashi 2002; O'Toole and King 2011). When participants' answers are evaluated, either exact scoring or semantic scoring can be applied. Exact scoring entails accepting only exact matches (bar minor spelling differences) with the original text as correct. In contrast, conceptual scoring, also called semantic scoring, makes it possible to accept an answer if it could be considered a synonym of the word from the original text. Based on the results of empirical studies, Jansen et al. (2017) conclude that the fixed-ratio method is preferable (see Bachman 1985), while exact scoring is most appropriate for determining the outcomes (see O'Toole & King 2011).

Criticism against readability formulas

Much of the criticism against readability formulas relates to the inability of formulas to help users create better texts (Klare 1984; Kleijn 2018). That is not the aim of a readability formula, however. Following Klare (1984), Kleijn (2018) distinguishes between prediction research and production research. While readability prediction is concerned with matching texts at a certain level with appropriate target audiences, traditionally by means of a readability formula based on correlations, readability improvement aims at guiding authors to produce better texts to reach their intended audience. Kleijn (2018) emphasises that, for prediction, it is not important to be able to explain why a text is appropriate for a reader or not, whereas in the case of text improvement causal explanation is the crux of the matter.

The use of traditional formulas for the prediction of readability has been criticised because the selection of linguistic variables relied heavily on how convenient it was to use them and less on their predictive value (François 2015; Schriver 2000). However, reader-specific and situational aspects, such as prior knowledge and motivation could also influence comprehension, as pointed out by DuBay (2004). Different formulas do not necessarily give the same results for the same texts, as different variables and different criteria have been

used in the development of the various formulas. Readability formulas can also sound more precise than they are (Duffelmeyer 1985).

Furthermore, a causal relationship between linguistic features and difficulty is often suggested when only a correlational relation has been determined (Jansen & Lenz 2008). Sentence length, for example, correlates with higher readability. However, if a writer were to manipulate their text to get a high readability score by indiscriminately shortening sentences, the text would not become more readable simply because it contains short sentences (Schriver 2000) – it might even become more difficult to understand.

Readability formulas for Afrikaans

There have been a few endeavours at devising readability formulas for Afrikaans, all but one based on the regression analysis approach. Van Rooyen (1986), Combrink (1992), McDermid Heyns (2007) and McKellar (2008) have all suggested formulas, yet only Van Rooyen has used real text comprehension data as part of the research to develop two formulas. Her formulas, however, are limited to texts intended for learners of 14 to 18 years (Van Rooyen 1986). Jansen et al. (2017) conducted a study to evaluate the formulas proposed by Van Rooyen (1986), McDermid Heyns (2007) and McKellar (2008) using cloze scores. None of those formulas proved to produce accurate readability predictions on novel texts, as we illustrate below.

In the study by Jansen et al. (2017), 10 texts written in Afrikaans were selected: 5 articles from a popular magazine and 5 documents used in governmental communication. All texts were about 300 words long. First, all linguistic features included in the four readability formulas[2] were measured for each text. To assess actual text comprehension, the cloze test was administered, using the fixed-ratio and exact scoring methods. For each text, 5 different cloze tests were developed. A total of 149 Afrikaans-speaking participants with different levels of education each completed a set of two of the 50 resulting cloze tests.

The quality of the formulas was determined by combining the predictions and the participants' cloze test scores. Both Van Rooyen formulas produced readability scores that were not significantly correlated with actual comprehension scores as measured by the cloze test (r=.34; p=.33, and r=-.21; p=.56). For the McKellar formula, the correlation with cloze

test scores was significant (r=.64; p=.04); for the McDermid Heyns formula, the correlation almost reached statistical significance (r=.61; p=.06) (Jansen et al. 2017: 149, 160).

The cloze test correlations of the formulas of McKellar (r=.64) and McDermid Heyns (r=.61) may seem adequate at first consideration. However, Jansen et al. (2017) point out that almost two thirds of the variance (R^2=.65) in the cloze scores could be predicted simply from average word length, with regression analyses showing that adding any other text characteristics already used in those formulas in addition to word length as predictors would not lead to stronger correlations.

Cornelius (2012; 2020) used Combrink (1992)'s *Misindeks* (based on the Gunning-Fog index, with adaptations for Afrikaans) in her research on plain language in Afrikaans, but considers the manual calculation it requires as a drawback. The Afrikaans adaptation differs from the original Gunning-Fog index in that the number of words with three or more syllables is to be divided by two, and not by three, as in the original index. Combrink (1992)'s rationale for the adaptation of the Gunning-Fog is that longer words in Afrikaans are more prevalent than in English, for which the Fog index has been developed originally, and that Afrikaans readers are therefore more used to navigating longer words than their English counterparts. Combrink (1992) does not provide any information on the validity of his adapted method, other than stating that he had tested it. After having used the *Misindeks* in their research, Du Plessis and Vos (2022) concur that more research is necessary to find an automated means of readability assessment for Afrikaans.

Given the above, we decided to explore a more user-friendly and accurate readability formula for Afrikaans. We wanted to include only linguistic features that could be measured in a fully automated manner to make it possible to build an automatic system. Only then can a formula for Afrikaans be developed that is on par with the applications that today's professional writers have at their disposal in, for example, English.

Method

We used a set of 40 Afrikaans texts from different genres and determined the readability of each of the 40 texts by means of cloze tests. In addition, we measured 88 linguistic features, filtered on the basis of readability prediction potential, of each of the 40 texts in an

automated manner (more details below). We then used the cloze scores as dependent variable and selected linguistic features to be used as predictors in a number of regression analyses. Below we elaborate on each of those steps.

Texts used for the cloze tests

The 40 Afrikaans texts that were used were each approximately 300 words long. The texts originated from 7 different genres: 6 articles from a popular weekly magazine, 6 texts produced by the South African government, 6 newspaper articles, 6 insurance brochures, 6 health brochures, 5 electronic newsletters and 5 informed consent texts. The set of 10 texts used in the study by Jansen et al. (2017) was included in the set of 40 texts for the current study.

Table 1 shows the number of texts from each genre. Permission was obtained to use texts that were not in the public domain, and specifically in the cases of the informed consent texts and e-newsletters, identifying details were replaced with pseudonyms or place-holder content.

Five cloze tests were created for each of the 40 texts, which yielded 200 hardcopy cloze tests.

Table 1: Number of texts and number of cloze tests per genre.

Genre	**Number of texts**	**Cloze tests created per genre**
Weekly magazine article	6	30
Government texts	6	30
Newspaper articles	6	30
Insurance brochures	6	30
Health brochures	6	30
Electronic newsletters	5	25
Informed consent texts	5	25
Total	**40**	**200**

The cloze tests were presented to 595 Afrikaans-speaking participants, who were each asked to complete 2 cloze tests, resulting in approximately 6 participants per cloze test, and hence approximately 30 participants per text. Participants always received cloze tests from two different genres, and they took part in the study only once. The collection of the data took place in three phases over four years, from 2016 to 2019.

Cloze scores were created by following the fixed-ratio method: every fifth word in each cloze test, with the exception of proper names and numbers, was deleted and replaced by a short line to indicate that a word was left out. All lines were of the same length. To ensure that each word of an original text would have an equal probability of being replaced by a line, in the first batch of cloze tests for each text, the first, sixth, eleventh, etcetera, word were deleted; in the second batch, the second, seventh, twelfth, etcetera, word were deleted, and so the process continued.

Participants

Data on participants' sex, age and highest level of qualification were collected. There were more female than male participants, the vast majority had completed at least high school, and the average age was 36 years. Most participants were optimistic about their reading abilities, judged by their responses to the DIALANG testing system statements[3] we asked them to consider. Only 4 out of 595 participants indicated a reading level lower than upper-intermediate level B2, indicating that, in general, our population would not have found the texts presented to them too challenging. (See participant details in Appendix 1.)

Data collection

The participants were approached by colleagues and students at Stellenbosch University in South Africa. All participants completed an informed consent form, which included information about the purpose, procedures, risks and benefits of the study. The form also reminded participants of their rights and the confidentiality of their responses.[4] After having given consent, each participant was presented with two cloze tests.

Participants were asked to use each gap in the cloze test to fill in the word they expected to have been in the original text. We followed the exact scoring approach to ensure consistent scoring. The cloze scores of the 200 cloze texts were then cleaned up. Gaps where nothing

was filled in were coded as 'empty', unless it seemed like a participant had not noticed uncompleted gaps in the last paragraph of a cloze test or realised that the cloze test continued on a second page (as the cloze tests were printed back-to-back if the whole text did not fit on one page). In such cases the gaps were coded as 'missing answer'.

If the total of missing answers for a certain cloze test of a certain participant was more than 30% of all gaps that could have been filled, the whole cloze test of that participant was disregarded. As a result, of the 1 190 cloze tests that were available, 25 cloze tests (2.1%) could not be used.

Linguistic analysis of the texts

We used the National Centre for Human Language Technology (NCHLT) annotation tools created for Afrikaans and made available by the Centre for Text Technology (CTexT) of North-West University in South Africa to automatically analyse the texts on which the cloze tests were based. These tools include the NCHLT Corpus software (Puttkammer & Schlemmer 2014a), NCHLT Tagger (Puttkammer & Schlemmer 2014b; SADiLaR Language Resource Repository n.d.) and NCHLT Afrikaans Lemmatiser (Puttkammer et al. 2014). Publications paving the way for these tools were Pilon (2005), Puttkammer and Van Huyssteen (2006) and Eiselen and Puttkammer (2014).

Among others, a frequency list for Afrikaans, a part-of-speech tagger, a named entity recognition tagger and a phrase chunk tagger were used to single out 88 linguistic features to consider as independent variables. Some of these variables have previously demonstrated their value for readability assessment, while others were anticipated to be potentially good indicators of readability for Afrikaans. The identified features included phrase chunks such as noun phrases, verb phrases and adverbial phrases; parts of speech such as nouns, verbs, pronouns and proper nouns identifying, for example, a person or a location; punctuation, data on word frequency identifying the words that occur in the top 1 000 words of a frequency list, and raw text counts such as number of sentences, number of words and number of characters. The various phrases, parts of speech, proper nouns and word frequency counts were all calculated per 100 words. In addition, there are features such as number of words per sentence and number of characters per word.

Statistical analysis

In order to determine cloze test scores, we disregarded missing answers but took gaps into account:

cloze test score = (number of correct answers) / (number of correct answers + number of incorrect answers + number of gaps)

Next, the mean cloze test score for each text was calculated, which was considered a measure of the readability of the original text. The statistical package IBM SPSS Statistics (version 29) was used for calculating the mean cloze scores and performing the regression analyses.

To establish whether it would be possible to create a stable formula by means of regression analysis that would also generalise well on new, unseen data, we conducted four-fold cross validation multiple regression, as Crossley et al. (2017) and Kleijn (2018) have done in the development of readability formulas by means of regression.[5] As Brownlee (2023) points out, K-fold cross validation works well for small datasets where one does not have a large amount of unseen data to test on. We split our data into 4 folds of 10, and then grouped 3 folds together in subsets to use for training, keeping the last fold out to test separately. This procedure was performed four times, thereby ensuring that each fold of 10 was used only once as a test fold. Four formulas were developed this way, and each time the remaining subset of 10 texts was used to evaluate the new formula's predictions.

Our selection criteria for predictors were similar to those that Kleijn (2018: 148) and Pander Maat et al. (2023: 12) applied in their endeavours to develop readability formulas for Dutch. We initially selected linguistic features as predictors for regression analysis only if the correlation with the mean cloze score indicated a consistent relationship: the correlation needed to be below -.40 or above .40. (We did also investigate correlations with the mean cloze score that were below -.30 or above .30, as a general guideline for correlation thresholds, but the cross-validation for the -.40/.40 threshold as used by Kleijn (2018) and Pander Maat et al. (2023) yielded more stable results.) From this subset, predictors were further selected only if their mutual correlation was between -.70 and .70. Mutual correlations beyond -.70 or above .70 would result in multicollinearity. When linguistic features did show mutual correlations stronger than .70 or weaker than -.70, their

correlations with the cloze score were compared. The linguistic feature with the strongest correlation with the mean cloze score was then included as a predictor – just as Kleijn (2018) and Pander Maat et al. (2023) did. As a last step, we selected only features that occurred in all 40 texts. This was done to prevent features that did not appear in some of the texts at all from having a disproportionate influence on the resulting formulas (compare Pander Maat et al. 2023, who also removed some features that did not occur in the data in a regular fashion for similar reasons).[6] We subsequently applied the selection criteria for predictors separately for each of the four subsets of 3 folds, which yielded the features listed in Table 2 to use as possible predictors for each fold. The different features that were available for each fold already indicated that the cross-validation formulas might differ from each other substantially. The feature abbreviations and their full forms (from Pilon 2005) are listed in Table 3.

Table 2: Possible predictors available for regression analysis after selection criteria have been applied

Subset	Possible predictors
1	characters_per_word, B-VERB, VTHOK
2	B-VERB, freq1000, NA
3	syllables_total, ASA, B-VERB, LB, NA, VTHOG
4	characters_per_word, ASA, B-VERB, NA

Table 3: Feature abbreviations and their full forms (Pilon 2005)

Feature abbreviation	Full form
ASA:	number of adjectives, positive degree, attributive per hundred words
B-VERB:	number of verb phrases per hundred words
characters_per_word:	number of characters the word consists of (word length)
freq1000:	number of words in the 1000 most common Afrikaans words provided by Eiselen and Puttkammer (2014)
LB:	number of definite articles per hundred words

syllables_total:	number of syllables in the whole piece
NA:	number of abstract nouns per hundred words
VTHOG:	number of unmarked, main, inseparable, transitive verbs
VTHOK:	number of unmarked, main, inseparable, copulative verbs

Results

Backward stepwise statistical regression analyses[7] were performed for each training subset of 3 folds, with the mean cloze scores for the 30 training texts as the dependent variable and the selected linguistic features (as listed in Table 2) as possible predictors. For subset 1 and 2, the analyses yielded only one formula incorporating all the predictors that were entered at the start of the analysis by means of the enter method, with no possibility of progressing to the backward stepwise method to improve on the initial model by discarding some predictors. The correlations between the predictions of the formulas for each subset and the cloze test scores found for the subsets of 30 training and 10 test texts can be seen in Table 4.

Table 4: Formulas with *r*, p and R^2 values obtained in four-fold cross-validation

Subset	Formula	3/4 training subset (30 texts)			1/4 test subset (10 texts)		
		r	**p**	R^2	***r***	**p**	R^2
1	47.739 – 4.594*characters_per_word + 1.196*BVERB + 1.954*VTHOK	.73	<.001	.54	.54	.88	.29
2	64.199 + 1.076*BVERB – 0.441*freq1000 – 1.728*NA	.76	<.001	.57	.42	.23	.17
3	57.390 – 0.033*syllables_total – 1.913*NA + 3.152*VTHOG	.72	<.001	.51	.18	.62	.03
4	87.435 – 6.731*characters_per_word – 1.526*NA	.62	<.001	.39	.55	.10	.30
	MEAN	**.71**		**.50**	**.42**		**.18**

There was a significant but moderate correlation between the mean cloze scores for each training subset and the outcomes of the respective formulas for each of those subsets (mean correlation for all the training subsets *r*=.71). The mean *R* squared value for the four training subsets was .50, indicating that, on average, 50% of the variance in the cloze scores could be accounted for by the predictors included in the formulas. The formulas performed worse on the test subsets: no significant correlations were found between the mean cloze scores for the test subsets and the outcomes of the respective formulas. As shown in Table 4, the mean correlation for all training subsets was .42, resulting in an *R* squared value for all

the test subsets of .18. This indicates that, on average, only 18% of the variance in cloze scores could be accounted for by the predictors included in the formulas.

Discussion

The current study has resulted in four possible readability formulas for Afrikaans. However, cross-validation showed that the outcomes of the traditional method of regression analysis that we applied to create readability formulas with the current data were not stable enough for any of these formulas to be considered as a useful instrument for writers or editors of texts in Afrikaans. The difference between the results for the cross-validation training subsets (mean R^2=.50) and their counterpart test subsets (mean R^2=.18) suggests that the developed formulas do not generalise well when used on new, unseen data. The overfitting of the model can most probably be attributed to the relatively small amount of data available for the analyses, and the possibility that the linear regression approach that was used is not powerful enough. Although no final readability formula can be put forward, the similarities in the cross-validation formulas may be of interest. Both the linguistic features *characters per word* (in two formulas) and *number of abstract nouns per hundred words* (NA) (in three formulas) showed a negative correlation with readability, whereas *number of verb phrases per hundred words* (B-VERB) (in two formulas) showed a positive correlation.

Word length, measured as characters per word or syllables per word, is a well-known readability predictor in traditional readability formulas (Flesch 1948; Gunning 1952; Kincaid et al. 1975). In general, the more frequent a word, the shorter it is. Another general observation holds that the shorter a word, the easier it is to understand – researchers of reading have indeed found that longer words take more effort to process as they "encode more linguistic units (visual features, characters, sounds, or morphemes)" (Kuperman et al. 2024). There would always be exceptions to this general observation – Bailin and Grafstein (2001) argue that there are many instances where short words can be more unfamiliar than longer words, and that more complex words may be more accessible to a reader precisely because of its seeming complexity: affixation and inflection, for example, may help language users to derive meaning, while making a word longer. This may be true, but it does not change the overarching predictive value of word length for readability assessment. Newer readability assessment models increasingly leverage word length alongside advanced

computational features to enhance readability assessment (see, for example, François & Bernhard 2014; Crossley et al. 2019).

The prevalence of abstract nouns emerging as a possible readability predictor is an outcome in line with research regarding the differences in processing abstract concepts and concrete concepts and words (see Paivio 1991; Schwanenflugel et al. 1992; Sadoski et al. 2000; Barsalou and Wiemer-Hastings 2005; Löhr 2023). According to Wiemer-Hastings and Xu (2005: 720) "it is more difficult to process abstract concepts such as *idea* than concrete concepts such as *table*. Such differences are known as concreteness effects, and span a variety of tasks, including studies of learning, memory retrieval, comprehension, lexical decision, translation, and semantic deficits." The frequency of abstract words plays an important role in the readability formulas recently developed for Dutch by Kleijn (2018) and Pander Maat et al. (2023). In their studies, concreteness (in contrast to abstractness) proved to contribute positively to comprehension. In a study by Kyle and Crossley (2015), their Tool for the Automatic Analysis of Lexical Sophistication also incorporates concreteness as a psycholinguistic word norm in measuring second-language lexical and speaking proficiency. While that tool is not aimed at assessing readability as such, its inclusion as a category in a tool of this nature underscores the role of concreteness in text readability.

The third predictor of readability that stood out in our analysis was the *verb phrase* (B-VERB) feature. Research by Healy and Miller (1970) on the verb as main determinant of the sentence shows that removing the verb from a sentence influences the integrity of a sentence negatively. Drawing on a study by Barzilay and Lapata (2008) on representing and measuring local coherence, Pitler and Nenkova (2008) argue that although more verb phrases in a sentence may increase sentence complexity, it makes for more interesting reading, and therefore may improve readability when information is clustered together in the sentence. In our study, verb phrases are indeed positively correlated to readability. Other verb features in the data correlating strongly with verb phrases were *unmarked, main, inseparable, transitive verbs* (VTHOG) (r=.62), and *present-tense inseparable modal verbs* (VTUOM) (r=.50). A likely explanation for the correlations is that those verb features would be components of verb phrases. Verb phrases as readability predictors are readily incorporated in studies that draw on natural language processing, for example Schwarm and Ostendorf (2005)'s SVM classifiers used to count verb phrase occurrences to calculate

readability, Pitler and Nenkova (2008)'s subsequent research, and the Tool for the Automatic Analysis of Syntactic Sophistication and Complexity by Kyle (2015).

The collection and subsequent scoring of cloze test responses in the current study contributed to the creation of a high-value dataset containing manual annotation. A limitation of this study is the size of the dataset, however. Our dataset consisted of 40 texts, split four times in a 75/25% ratio, to yield 30 training texts and 10 test texts for four cross-validation subsets. Referring to Khamis & Kepler (2010), Tabachnik and Fidell (2013: 123) recommend $n>20+5m$ as the minimum sample size for multiple regression, where n is the sample size and m is the number of predictors. In the case of the current study, in which a maximum of five predictors were used in the cross-validation regression analyses leading to the formulas that were considered, this would require our training set to have contained more than 20+5*5=45 texts. Over 45*30 (the number of participants per text in this study)=1 350 participants would then have been required for the training set, plus over (15*30=) 450 participants needed to collect cloze scores from for the extra set of 15 test texts.[8] In a labour-intensive data collection approach such as the one taken in the current study, a much larger number of texts than that to which we had access is not feasible.

In the context of the rapid development of natural language processing, particularly regarding large language models, and the improvement of automatic readability assessment over the past 10 years, a change of course in the development of readability assessment for Afrikaans may well yield better results. The recent development of powerful pre-trained neural language models such as BERT (Devlin et al. 2019) has revolutionised the natural language processing arena. Martinc et al. (2021) and Vajjala (2022) have shown that pre-trained neural language models such as BERT can be used for readability assessment – without including linguistic features explicitly. Further training and finetuning pretrained Afrikaans language models on readability assessment tasks may be good avenues to explore for further research into Afrikaans readability assessment. The XLM-R language model (Conneau et al. 2020), a multilingual language model trained on 100 languages including Afrikaans, as well as the NCHLT Afrikaans RoBERTa language model (Eiselen n.d.) are examples of language models that may be suitable for such research. Established large language models such as the GPT models, LLaMA or Claude could also be investigated in this regard (see, for example, Lopes Rego et al. 2024).

Conclusion

This study aimed at developing a readability formula for Afrikaans, using an approach frequently applied by prominent traditional readability formulas.

The readability of 40 Afrikaans texts was established by means of cloze tests, and 88 linguistic features were extracted automatically from those same texts. Linear regression with four-fold cross-validation was performed using cloze scores as dependent variable and selected linguistic features as predictors to determine whether stable readability formulas could be developed. The difference between the results for the cross-validation training subsets of 30 texts each (variance explained: 50%) and their counterpart test subsets of 10 texts each (variance explained: 18%) suggests that the developed formulas will not generalise well when used on new, unseen data. The overfitting of the model may be attributed to the relatively small amount of data that was available for the analyses, and possibly limited power of the traditional type of model that was used. Our findings underscore the importance of testing newly developed readability formulas on independent datasets to determine generalisability before releasing them to be used.

Although no final readability formula can be put forward, the linguistic features emerging as the most frequent predictors in the cross-validation formulas may be useful for future research: Word length and concreteness, both with a strong negative correlation with readability, respectively occurred in two and three of the four formulas, whereas the number of verb phrases, occurring in two formulas, showed a strong positive correlation.

A high-value dataset with readability labels based on data from participants was created in the current study. The labour-intensive data collection approach, however, limited the size of the dataset, which, in turn, impacted negatively on our endeavours to develop a robust readability formula by means of the traditional statistical method. The linear regression method that we used, although having been used in the development of several readability formulas before, may also not have been the most suitable method. It may be beneficial to shift from the statistical method used in the current study to more powerful pretrained neural network approaches. Further training and finetuning pretrained Afrikaans language models on readability assessment tasks will be good avenues to explore for further research into readability assessment for Afrikaans.

Authorship contribution statement

Susan Lotz: Data collection, conceptualisation, methodology, formal analysis, writing – original draft, writing – review & editing. **Bo Blankers:** Data collection and processing, methodology, writing – review. **Rik van Noord:** Methodology, writing – review & editing, supervision. **Carel Jansen:** conceptualisation, data collection, methodology, assistance with formal analysis, verification, writing – review and editing, supervision.

Data availability

The data that support the findings of this study are available from the corresponding author, on reasonable request.

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Endnotes

[1] DuBay (2004) correctly points out that correlations play a much larger role than causal relations in readability research. A correlation coefficient (represented as *r*) is a value between 1 and -1 that statistically describes the relationship between two variables. A value of 1 or -1 indicates a perfect correlation, the + or - sign depending on whether the variables are positively or negatively related. A coefficient of 0 means there is no correlation, indicating no relationship. The square of the correlation coefficient (represented as R^2) indicates the percentage of the variance in scores for one variable that is accounted for by the variance in the scores of the other variable.

[2] The linguistic features included in the four already existing readability formulas were: syllables per sentence; concrete nouns per 100 words; infinitive constructions per 100 words; letters per sentence; words with more than three syllables per 100 words; pronouns per 100 words; words per sentence; syllables per word; percentage of words in list of 50 most frequently used words; pairs of brackets per 100 words; numbers and symbols per 100 words; and characters per word.

[3] We asked participants to self-assess their reading proficiency by responding to 17 so-called "Can Do" statements taken from DIALANG (N.d.). DIALANG is a language testing system that allows users

to perform a diagnostic test of their skills as second-language speakers in different languages, and relates the test results to CEFR (Common European Framework of Reference for Languages) levels (Alderson, 2005; Alderson and Huhta, 2005; see also Van Onna and Jansen, 2006). In the CEFR, six language proficiency levels are distinguished: beginner (A1), elementary (A2), intermediate (B1), upper-intermediate (B2), advanced (C1) and upper-advanced (C2). Since all participants in this study were native speakers, we used only the statements related to the B1, B2, C1 and C2 levels, as provided by the Council of Europe (2001:231).

[4] Permission for gathering the cloze test data had been granted by the Stellenbosch Research Ethics Committee: Human Research (Humanities) (Proposal number SU-HSD-002010, project number 4479) before data collection.

[5] Both Crossley et al. (2017) and Kleijn (2018) have used k-fold cross-validation in the development of readability formulas via regression. Crossley et al. (2017) developed 10 formulas in a ten-fold cross validation and averaged their results, whereas Kleijn (2018) followed a related but different approach, conducting a five-fold cross-validation to test a new readability measure for Dutch. In her study, one formula was developed in a training phase, after which this formula was tested in five validation folds.

[6] For machine-learning methods, this restriction would not have to be imposed.

[7] Backward stepwise regression is a type of stepwise regression analysis that involves starting off with a model that contains all the predictors. They are then deleted one by one if they do not contribute significantly to regression. (Field, 2009, pp. 212-213 and Tabachnik & Fidell, 2013, pp. 139-140). Field (2009: 213) recommends backward stepwise regression over forward stepwise regression, as "forward selection is more likely than backward elimination to exclude predictors involved in suppressor effects". Suppressor effects occur "when a predictor has a significant effect but only when another variable is held constant". Field (2009: 213) warns that when forward selection is used, one runs a higher risk of making a Type II error, which involves missing a predictor that does, in fact, predict the outcome.

[8] If 75% of the total dataset amounts to 45 texts (for training), the size of the total dataset required is (45/0.75)=60 texts, and 25% of this dataset would then equal 15 texts, the number of texts that would be required for testing. If one were to reason that there were originally 88 and not only 5 predictors, the 88 predictors would evidently require a much higher number of texts and participants.

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